

Distributed Systems Security

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Security Threats, Policies, and Mechanisms

- Types of security threats to consider:
- Interception
- Interruption
- Modification
- Fabrication

Example: The Globus Security Architecture (1)

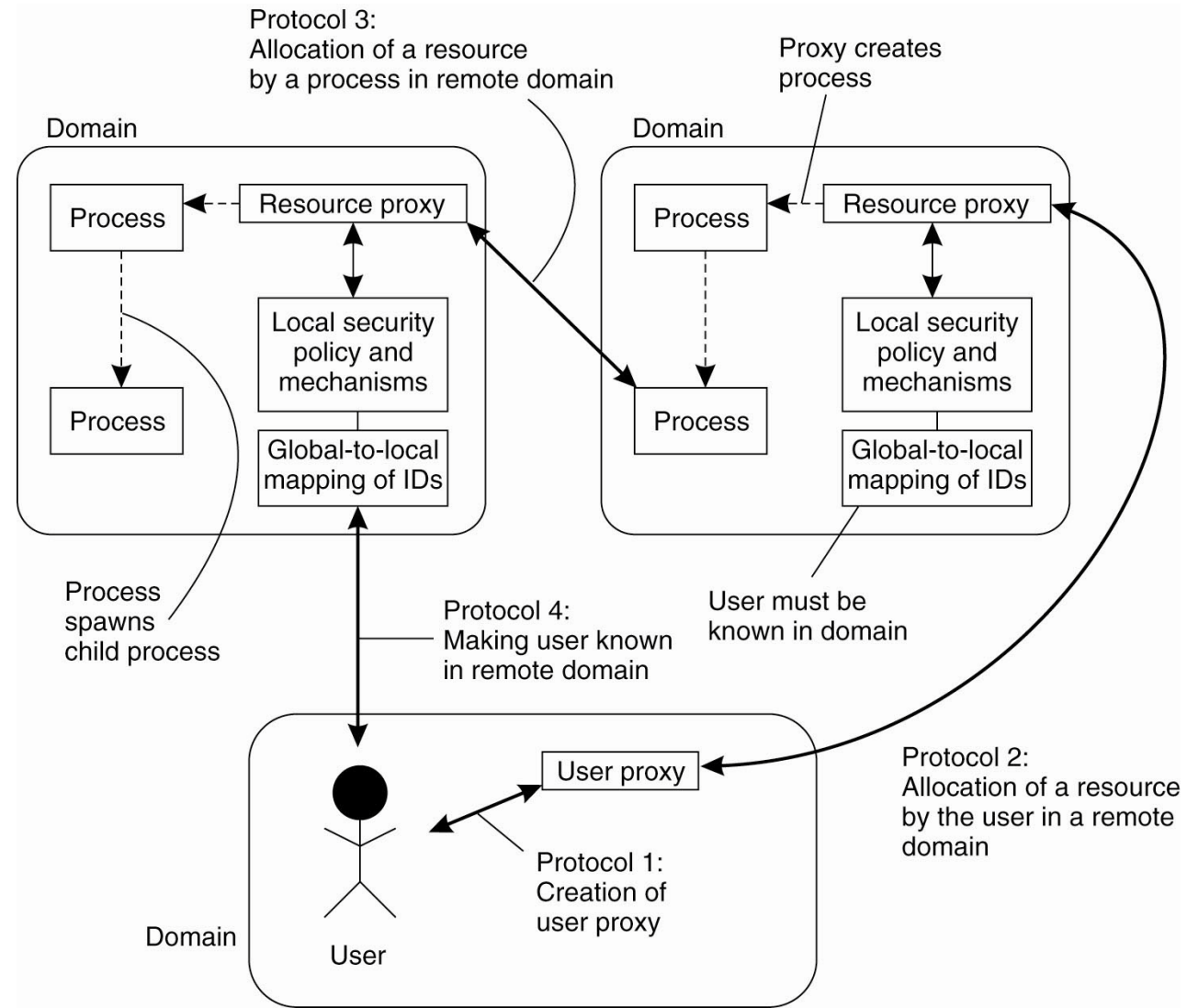
1. The environment consists of multiple administrative domains.
2. Local operations are subject to a local domain security policy only.
3. Global operations require the initiator to be known in each domain where the operation is carried out.

Example: The Globus Security Architecture (2)

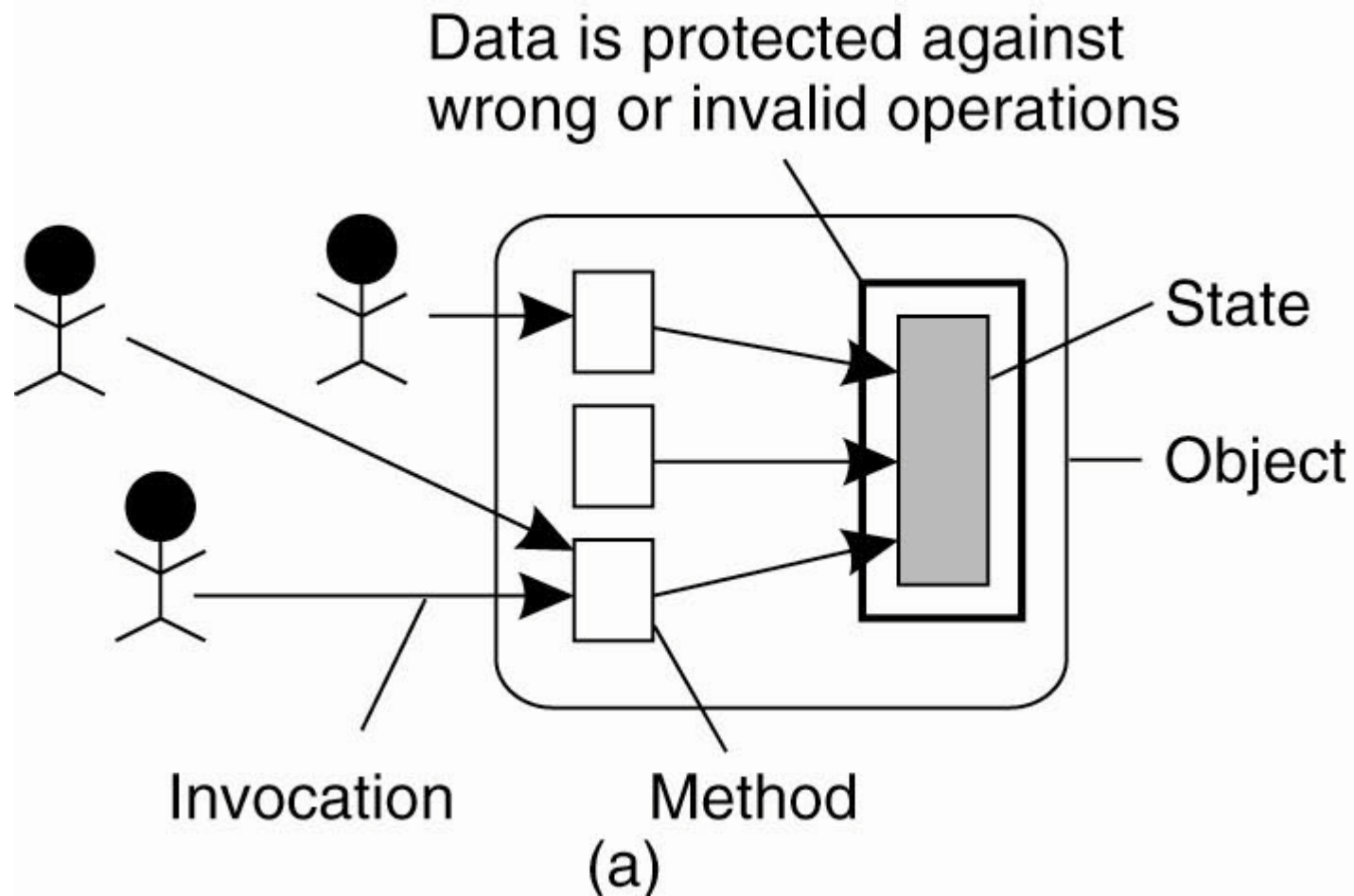
4. Operations between entities in different domains require mutual authentication.
5. Global authentication replaces local authentication.
6. Controlling access to resources is subject to local security only.
7. Users can delegate rights to processes.
8. A group of processes in the same domain can share credentials.

Example: The Globus Security Architecture (2)

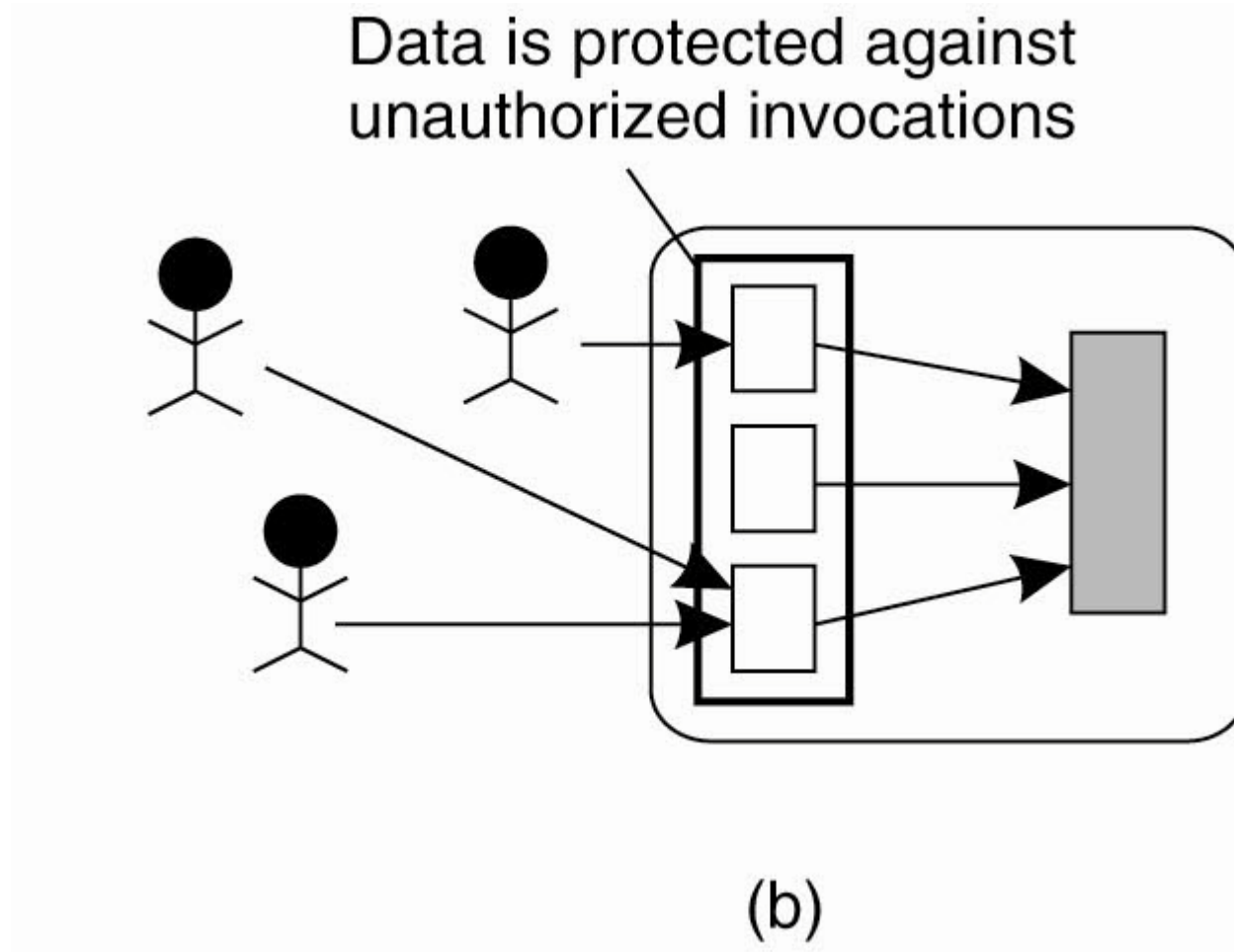
- The Globus security architecture.



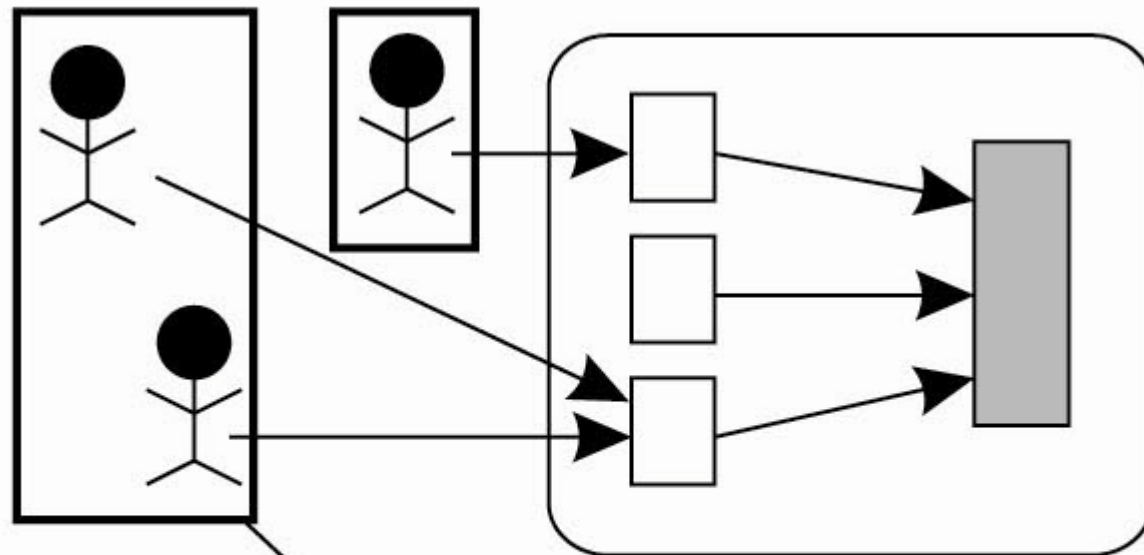
Focus of Control (1)



Focus of Control (2)



Focus of Control (3)

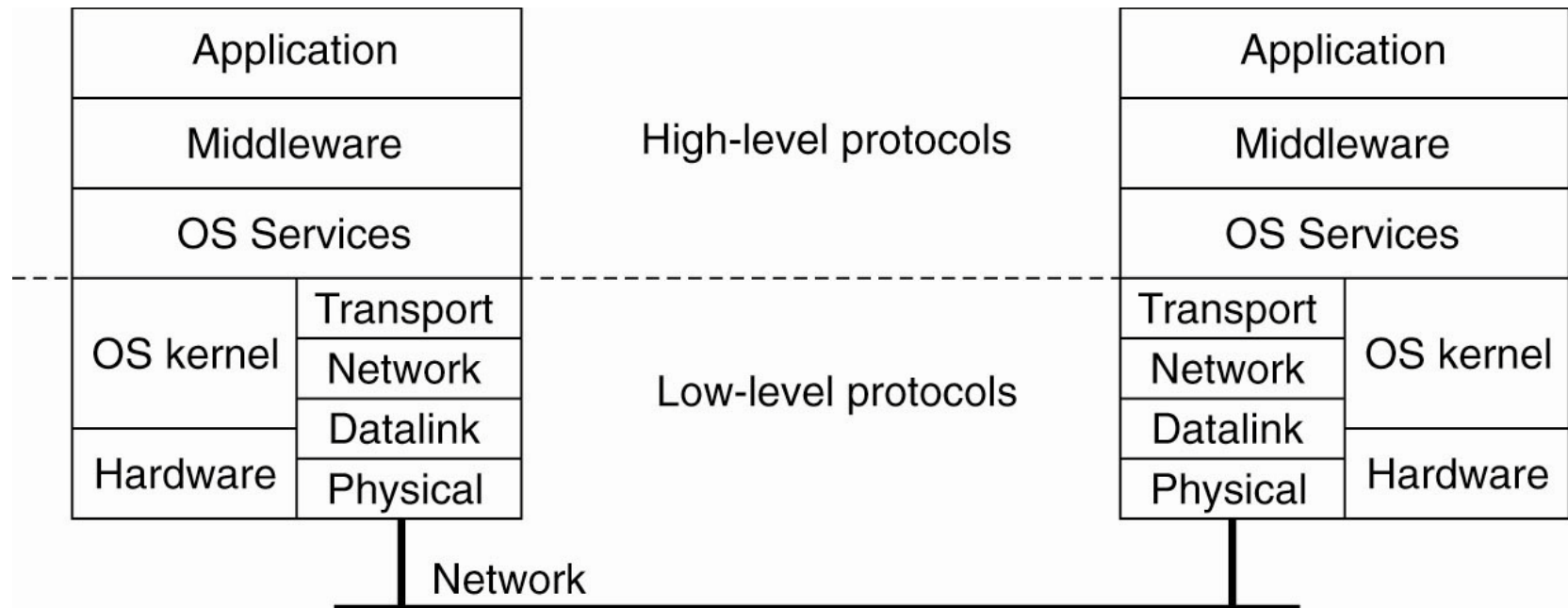


Data is protected by
checking the role of invoker

(c)

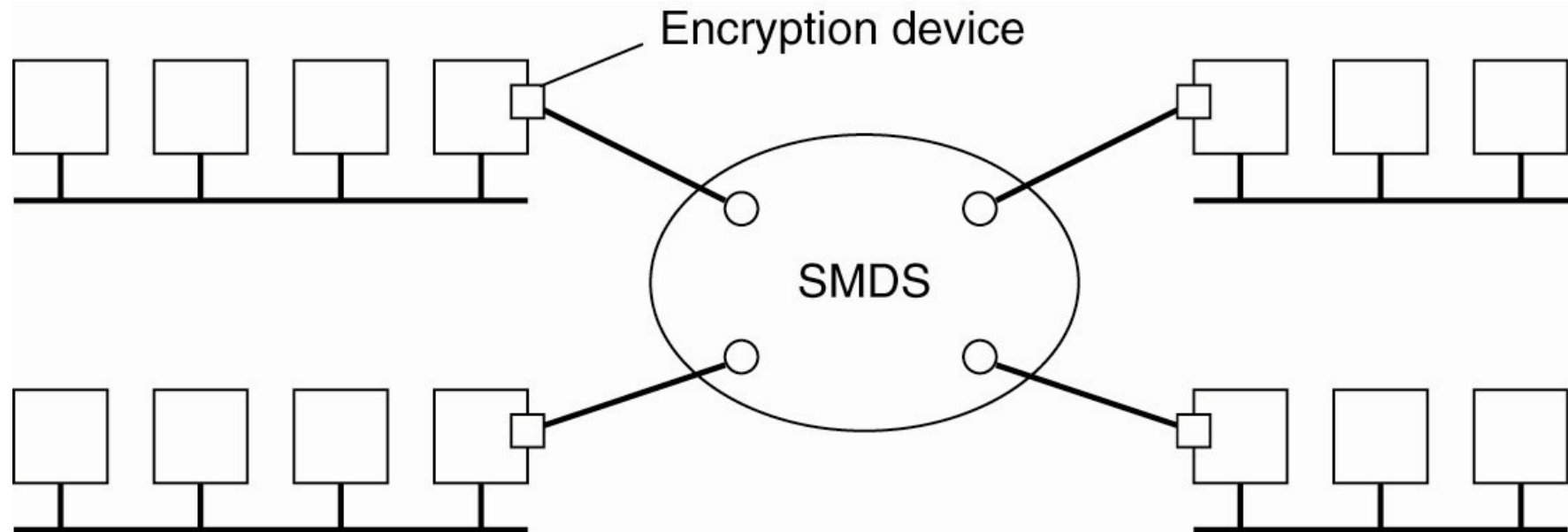
Layering of Security Mechanisms

(1)

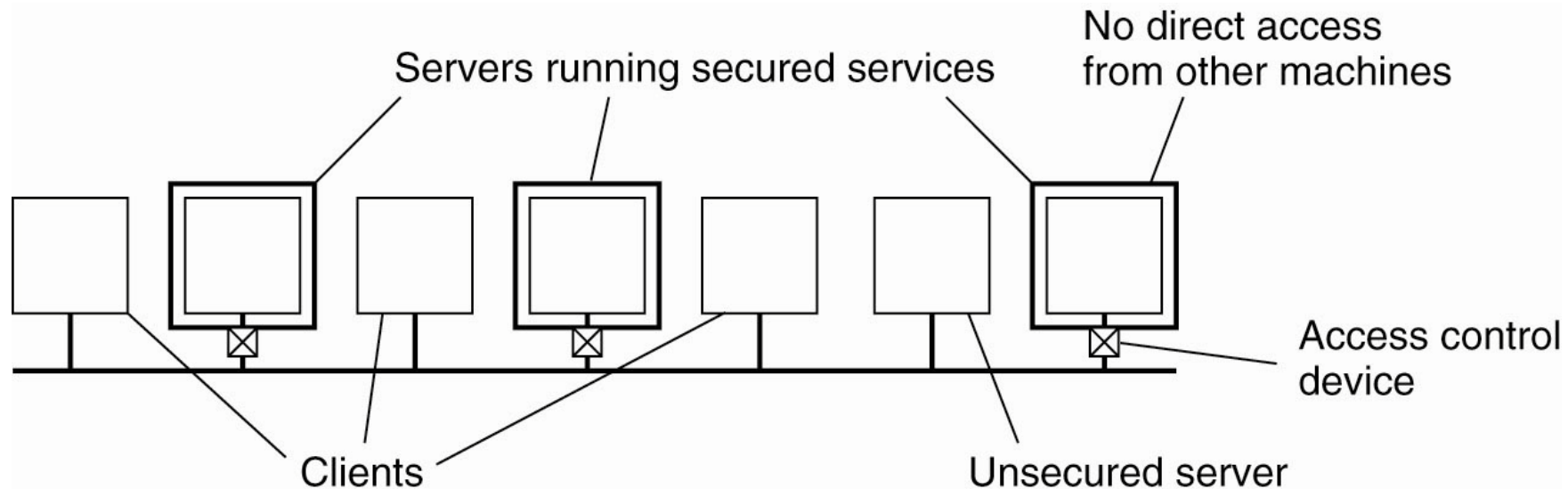


Layering of Security Mechanisms

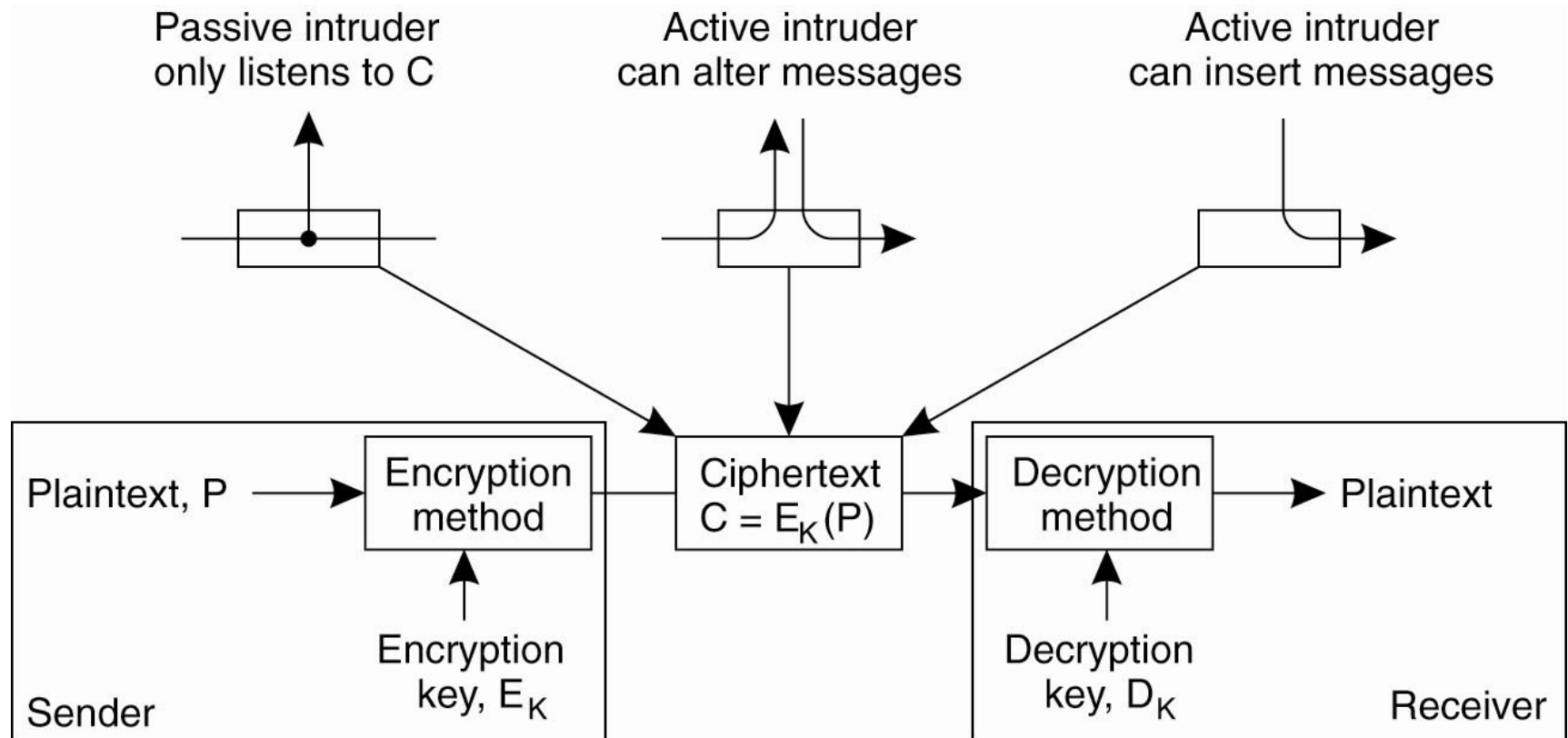
(2)



Distribution of Security Mechanisms



Cryptography (1)

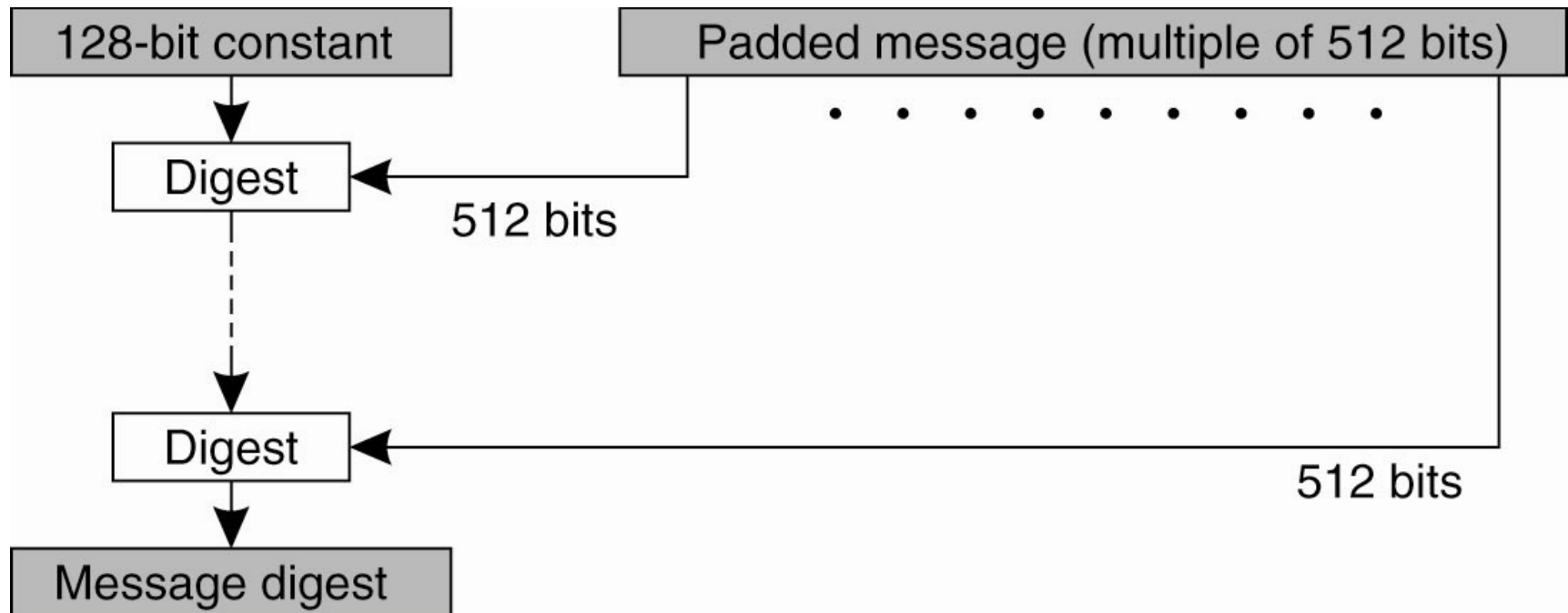


Cryptography (2)

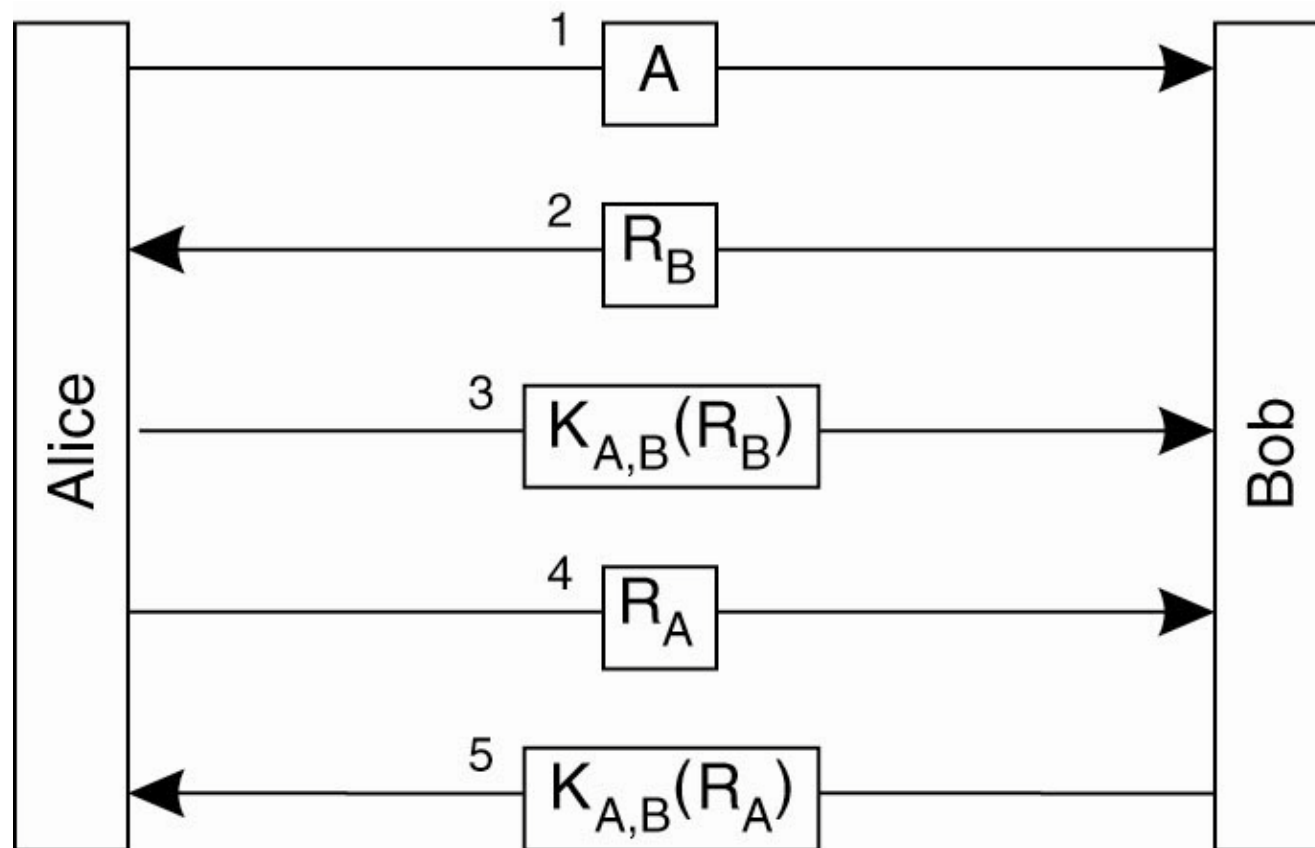
- Notation used

Notation	Description
$K_{A,B}$	Secret key shared by A and B
K_A^+	Public key of A
K_A^-	Private key of A

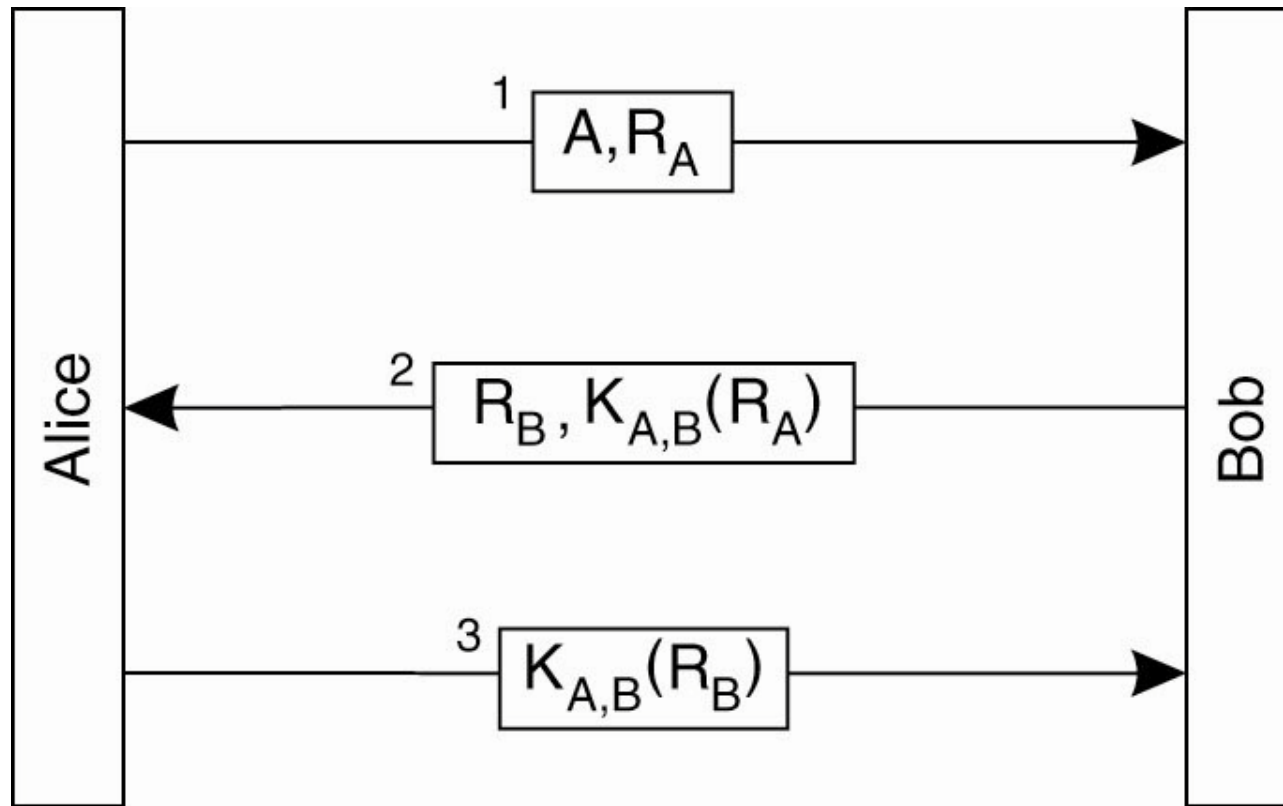
Hash Functions: MD5 (1)



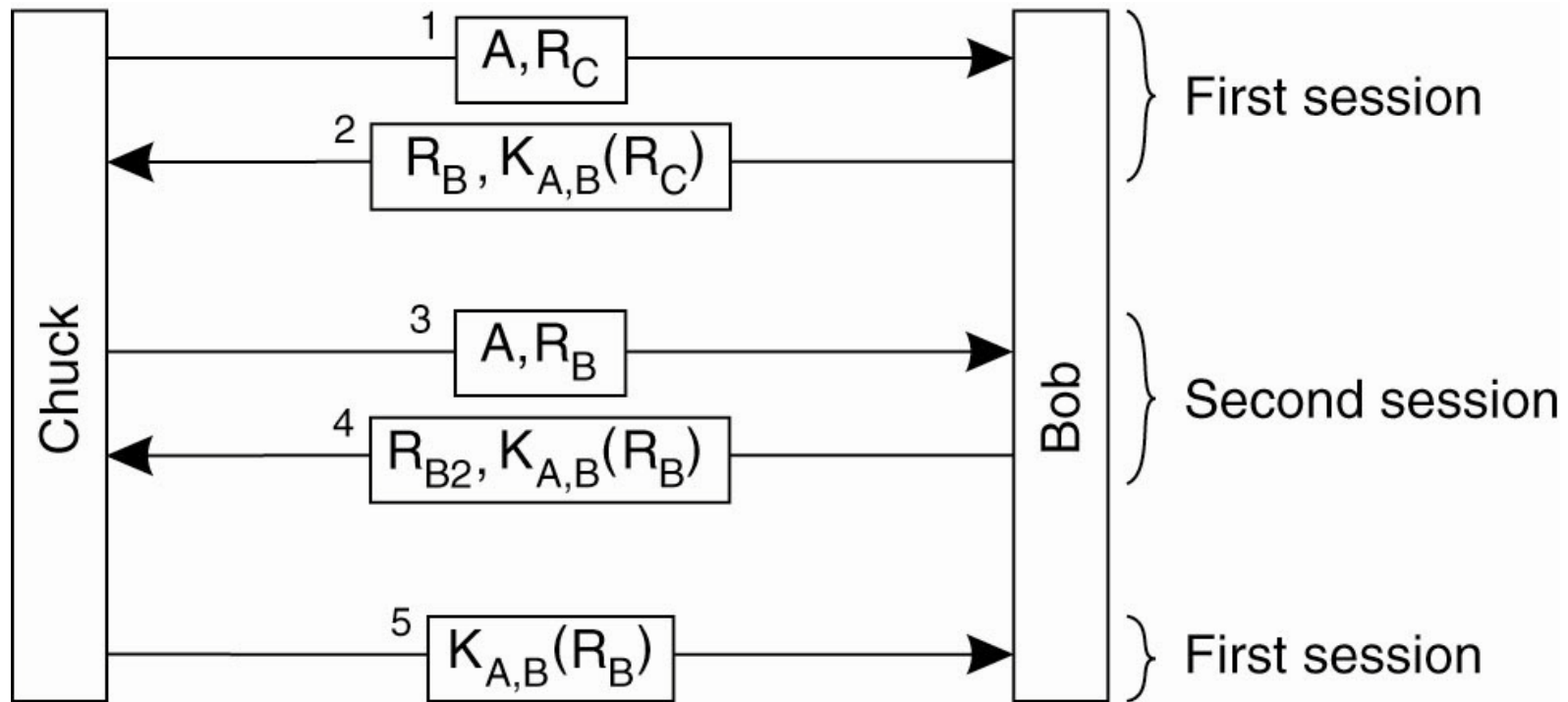
Authentication Based on a Shared Secret Key (1)



Authentication Based on a Shared Secret Key (2)

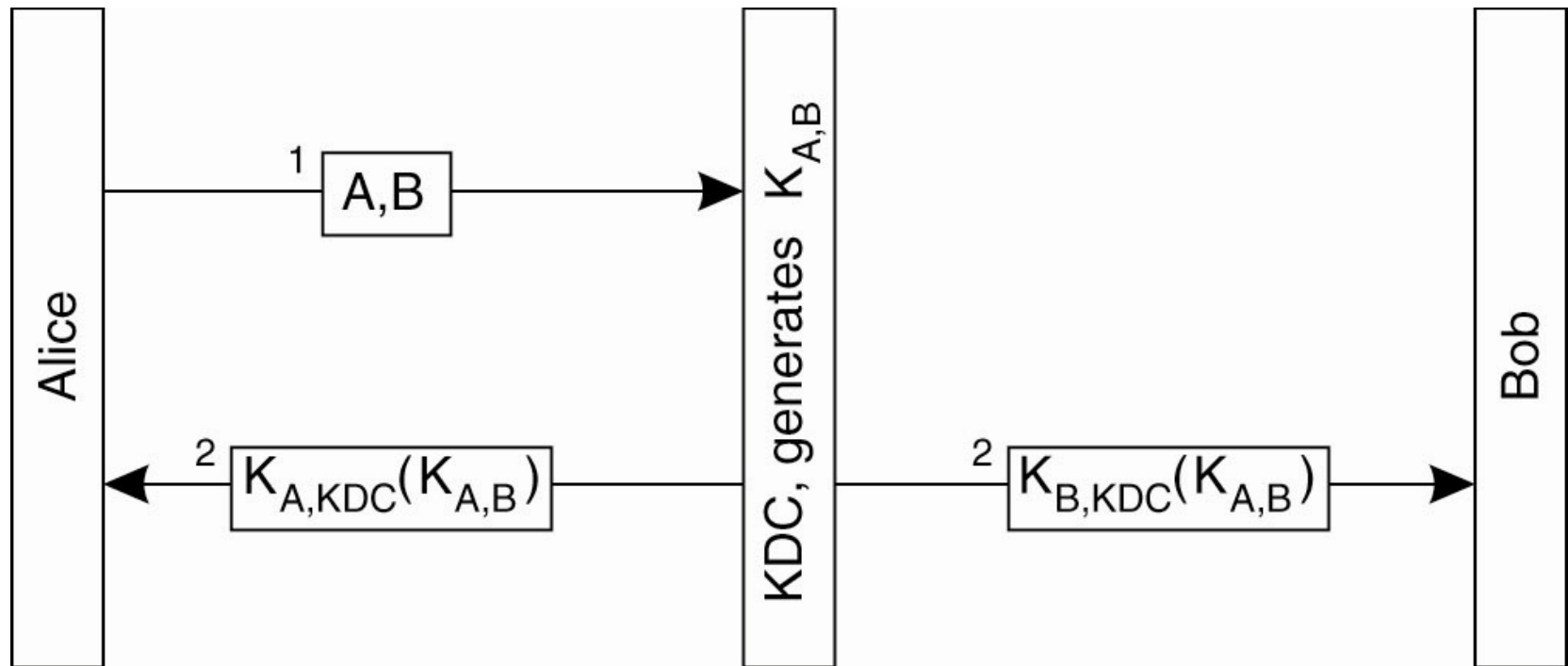


Authentication Based on a Shared Secret Key (3)

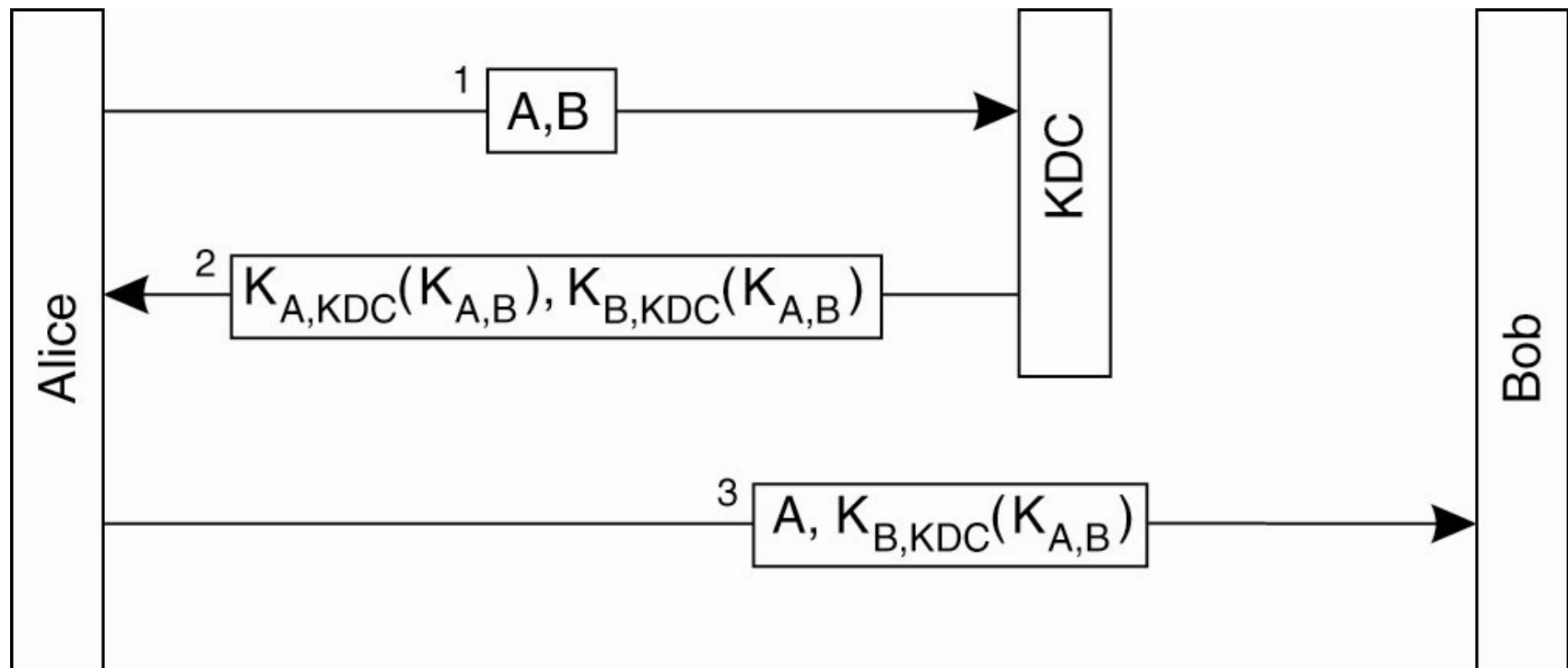


- The reflection attack.

Authentication Using a Key Distribution Center (1)

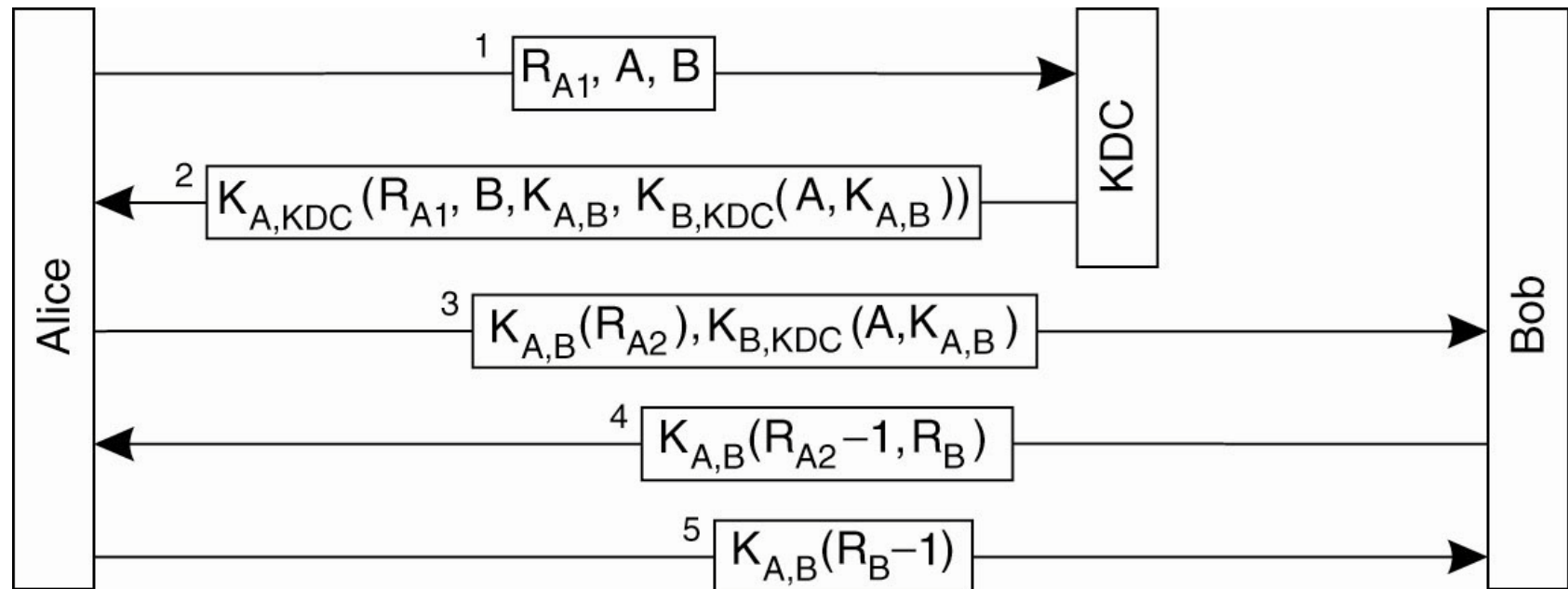


Authentication Using a Key Distribution Center (2)



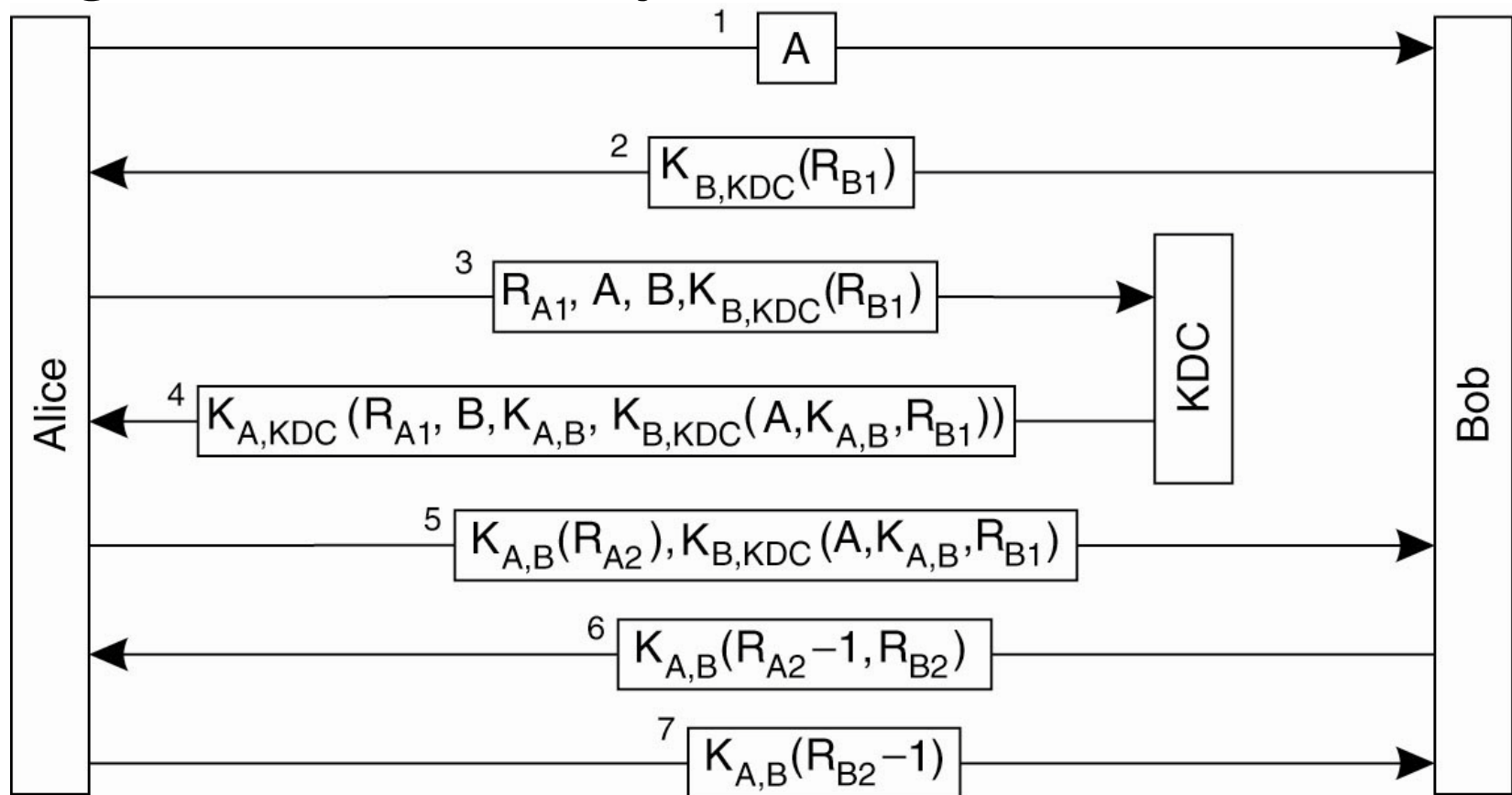
Authentication Using a Key Distribution Center (3)

- The Needham-Schroeder authentication protocol.



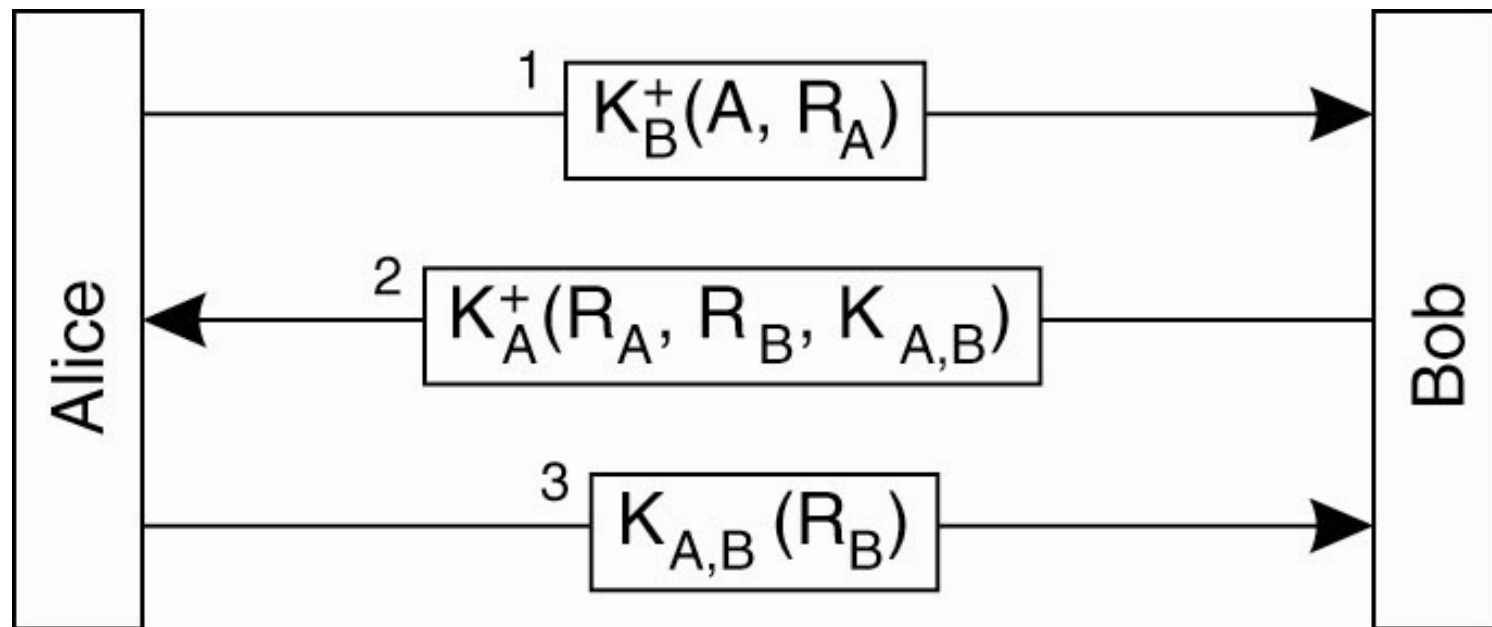
Authentication Using a Key Distribution Center (4)

- Protection against malicious reuse of a previously generated session key in the Needham-Schroeder

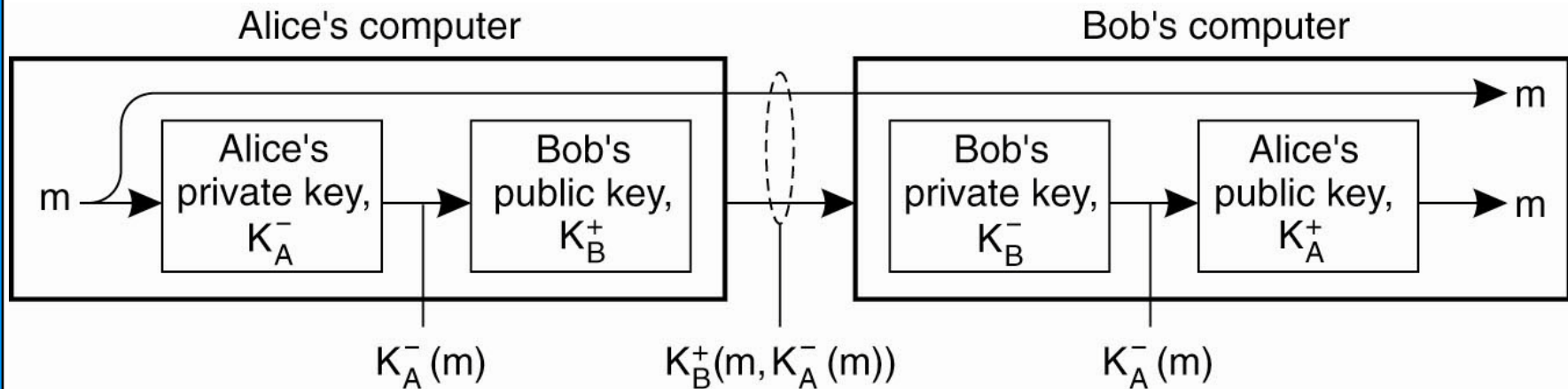


Authentication Using a Key Distribution Center (5)

- Mutual authentication in a public-key cryptosystem.



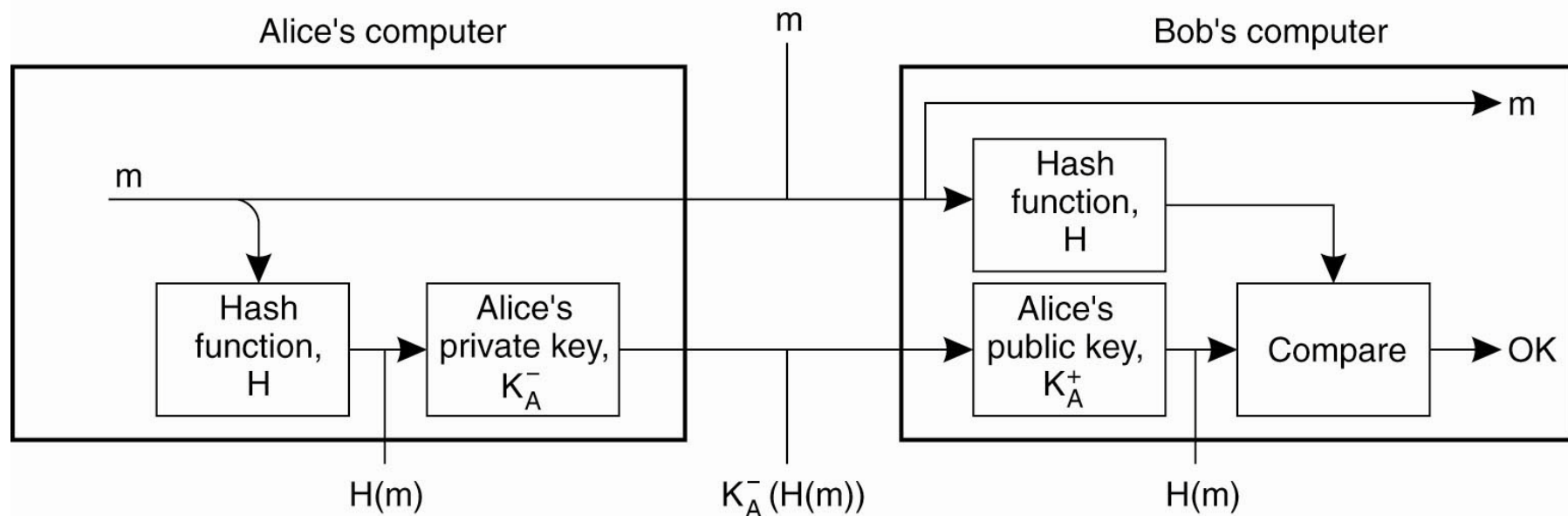
Digital Signatures (1)



- Digital signing a message using public-key cryptography.

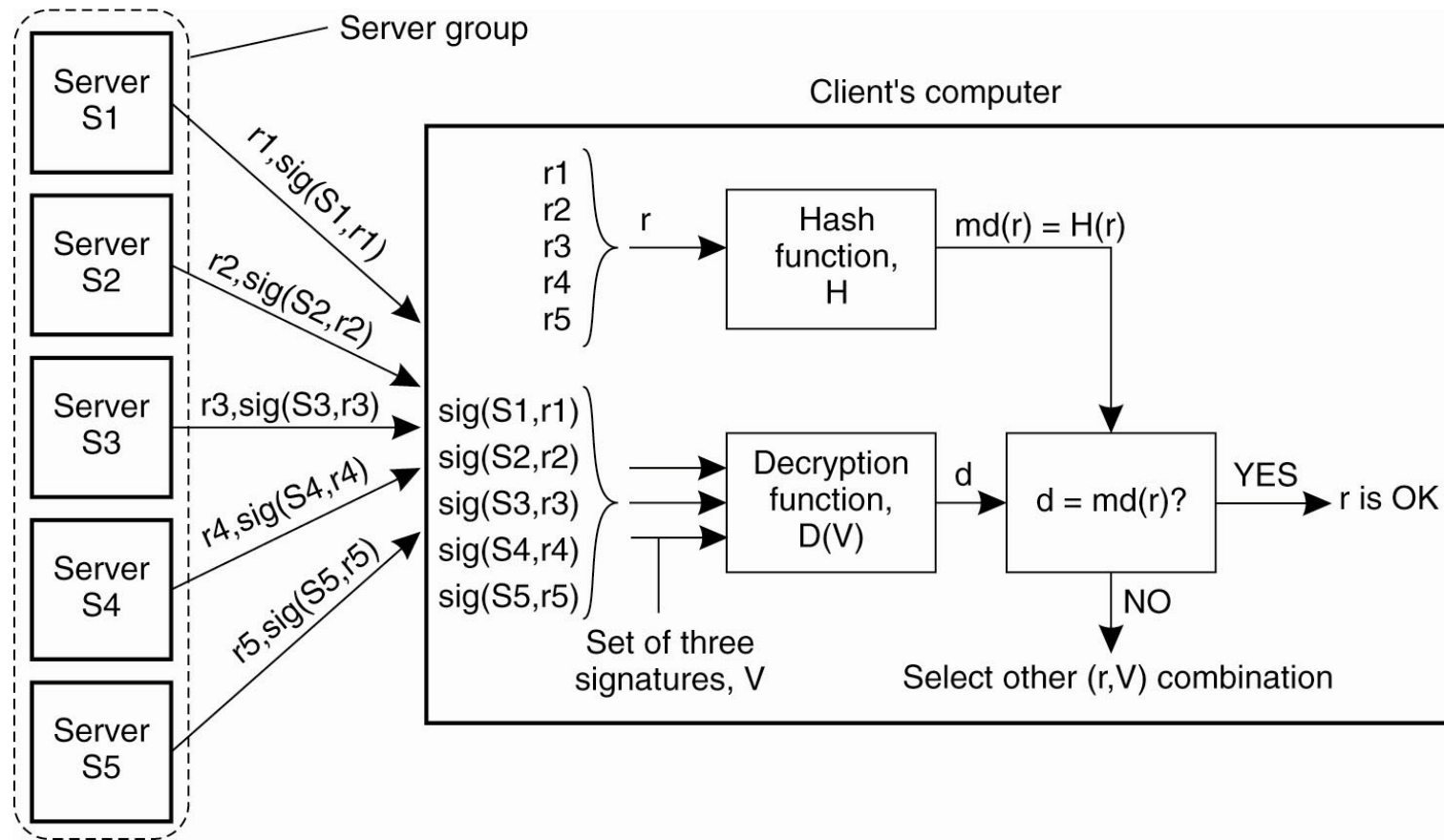
Digital Signatures (2)

- Digitally signing a message using a message digest.



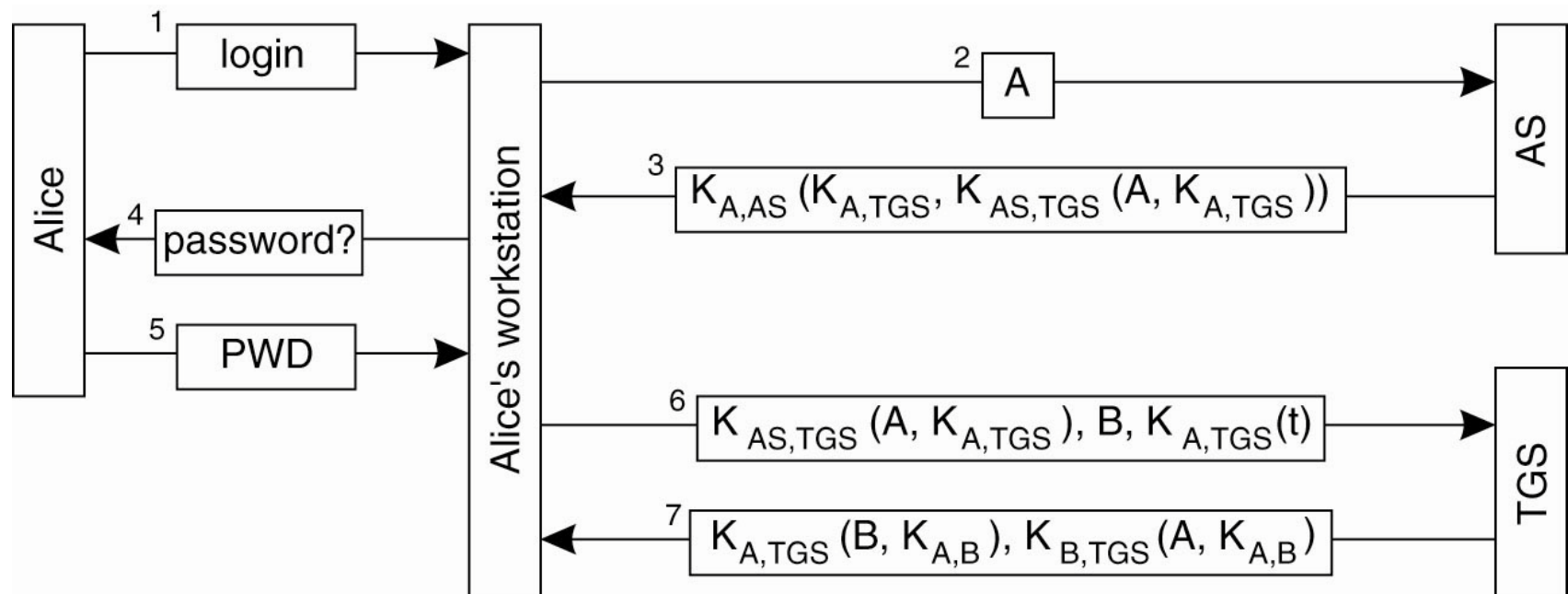
Secure Replicated Servers

- Sharing a secret signature in a group of replicated servers.



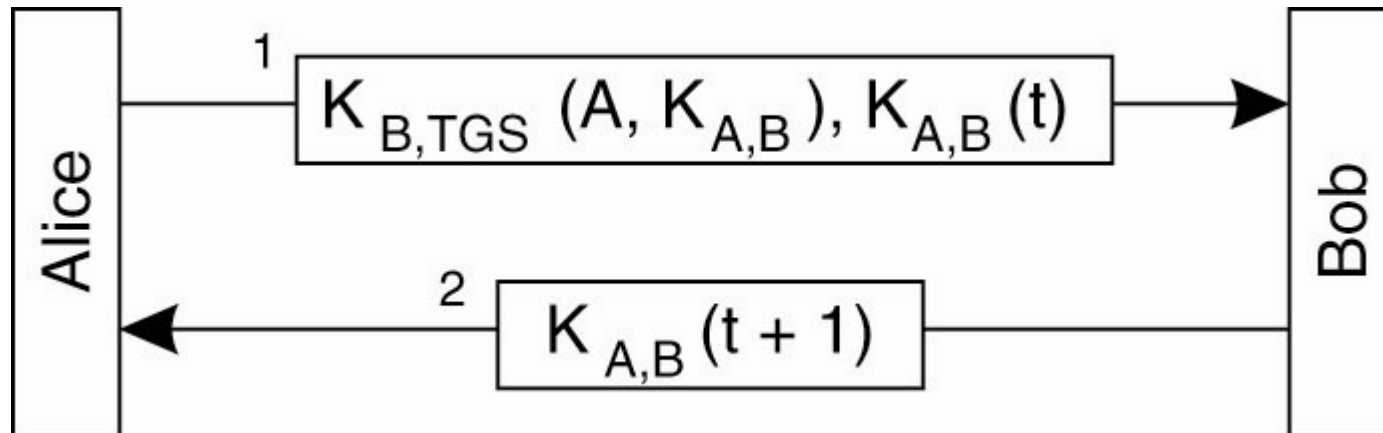
Example: Kerberos (1)

- Authentication in Kerberos.



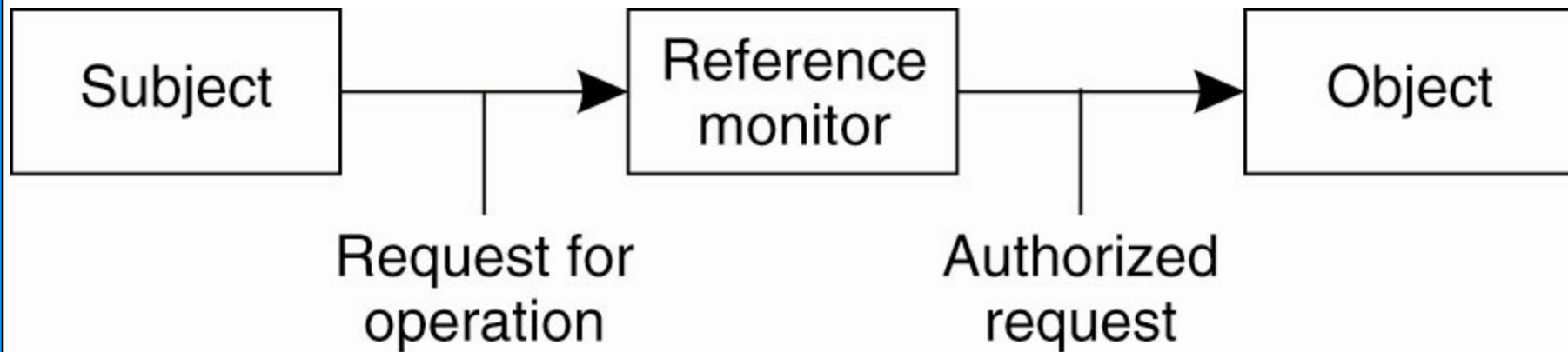
Example: Kerberos (2)

- Setting up a secure channel in Kerberos.



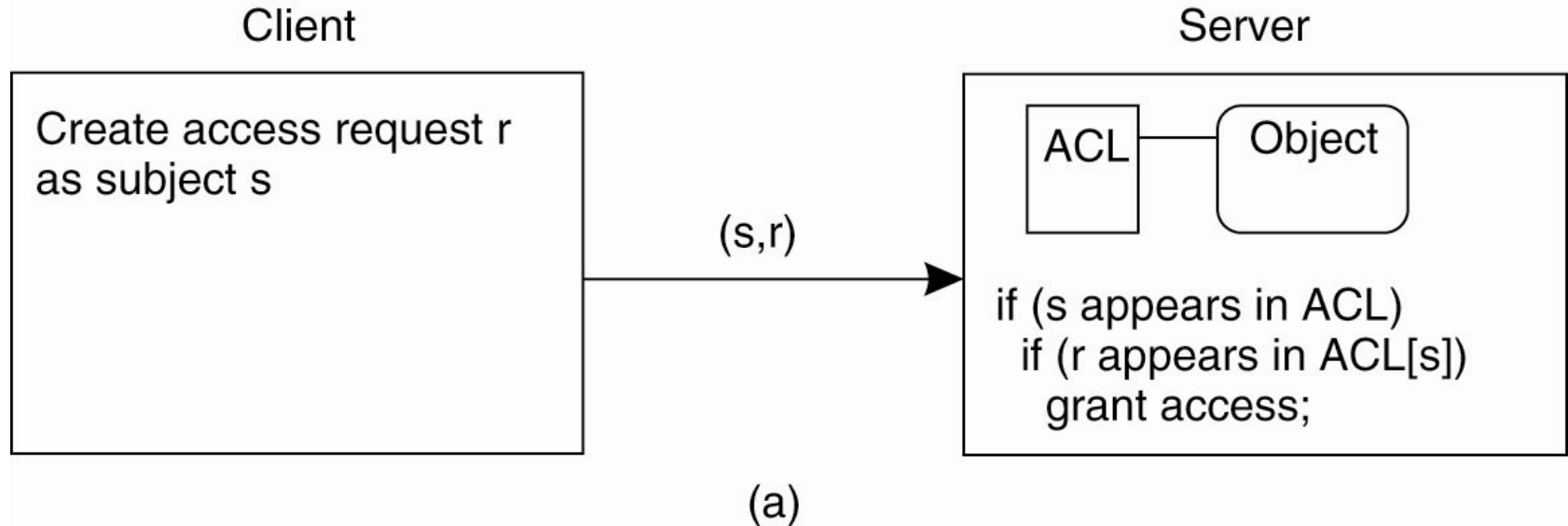
General Issues in Access Control

- General model of controlling access to objects.



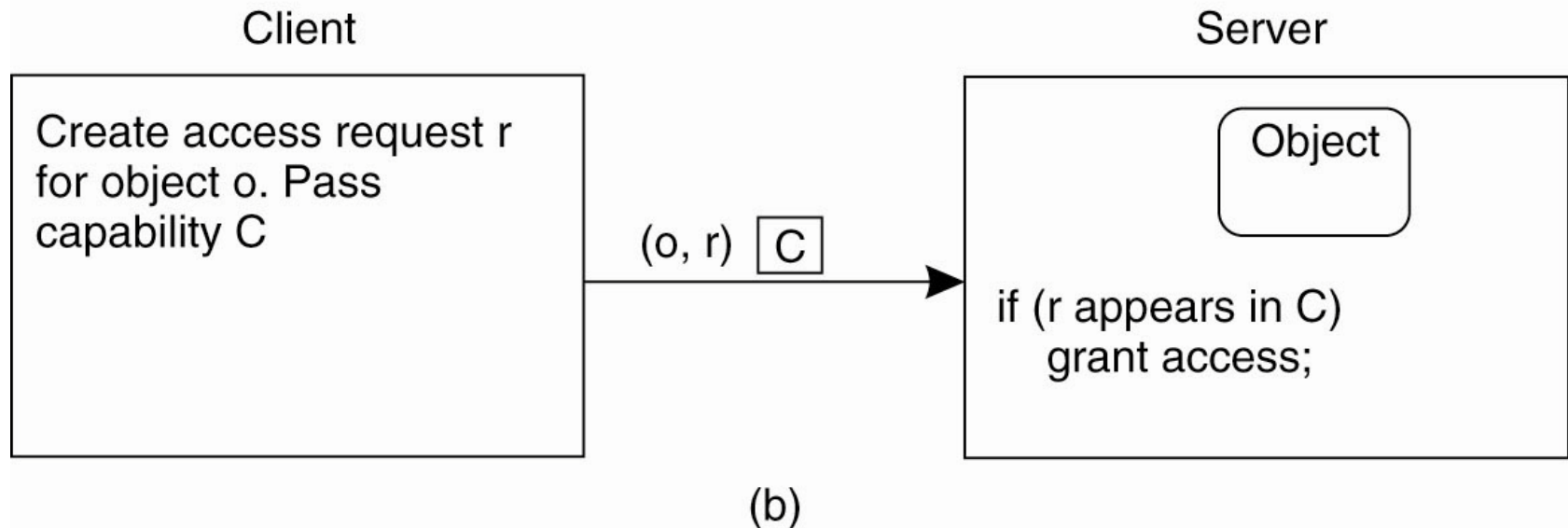
Access Control Matrix (1)

- Figure 9-26. Comparison between ACLs and capabilities for protecting objects. (a) Using an ACL.



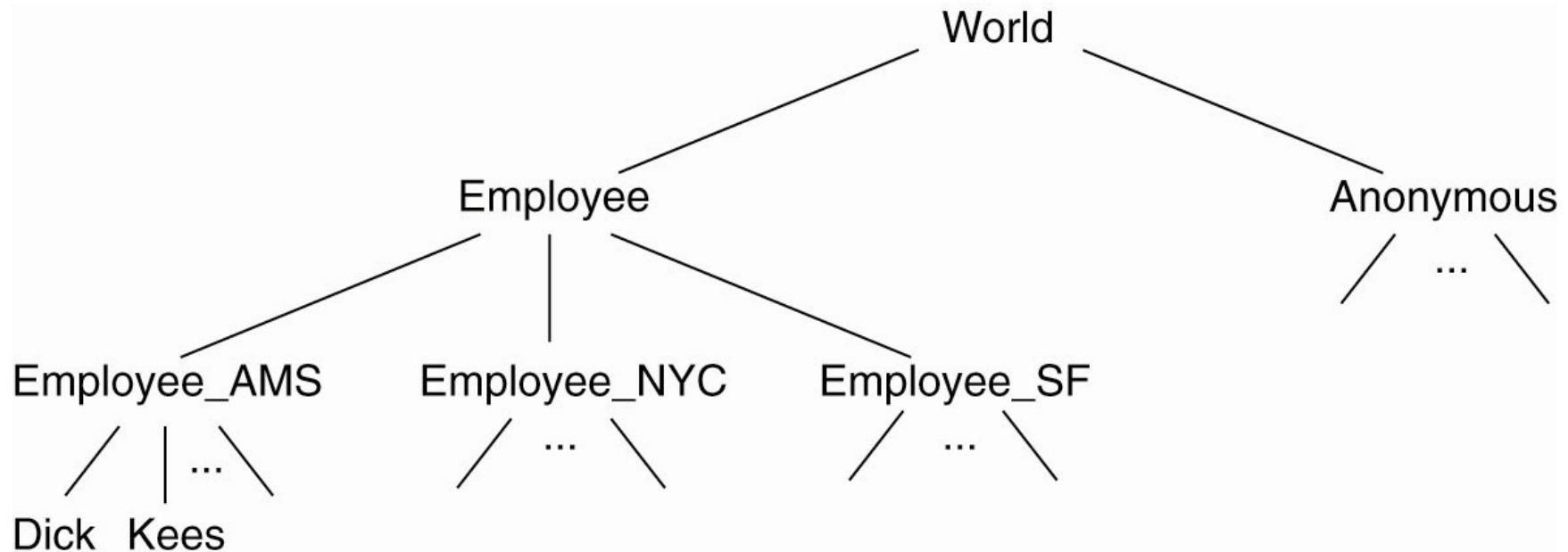
Access Control Matrix (2)

- Comparison between ACLs and capabilities for protecting objects. (b) Using capabilities.

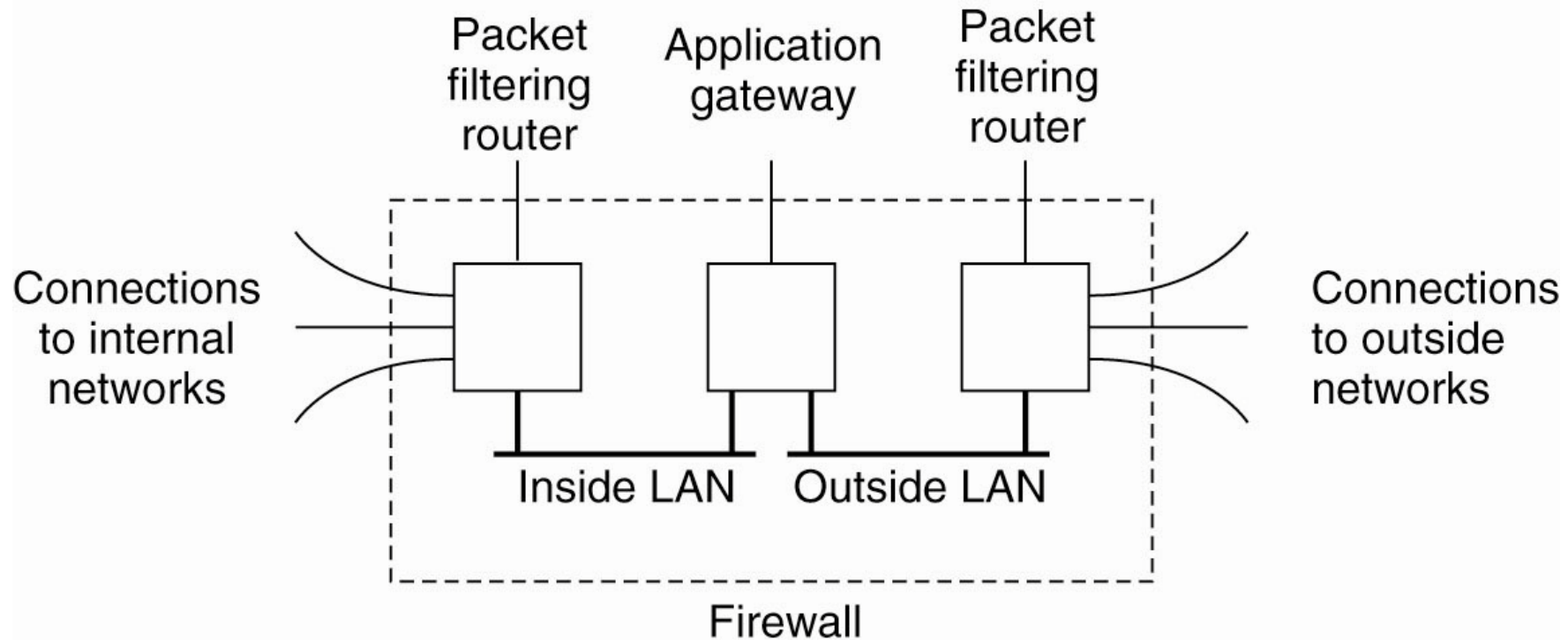


Protection Domains

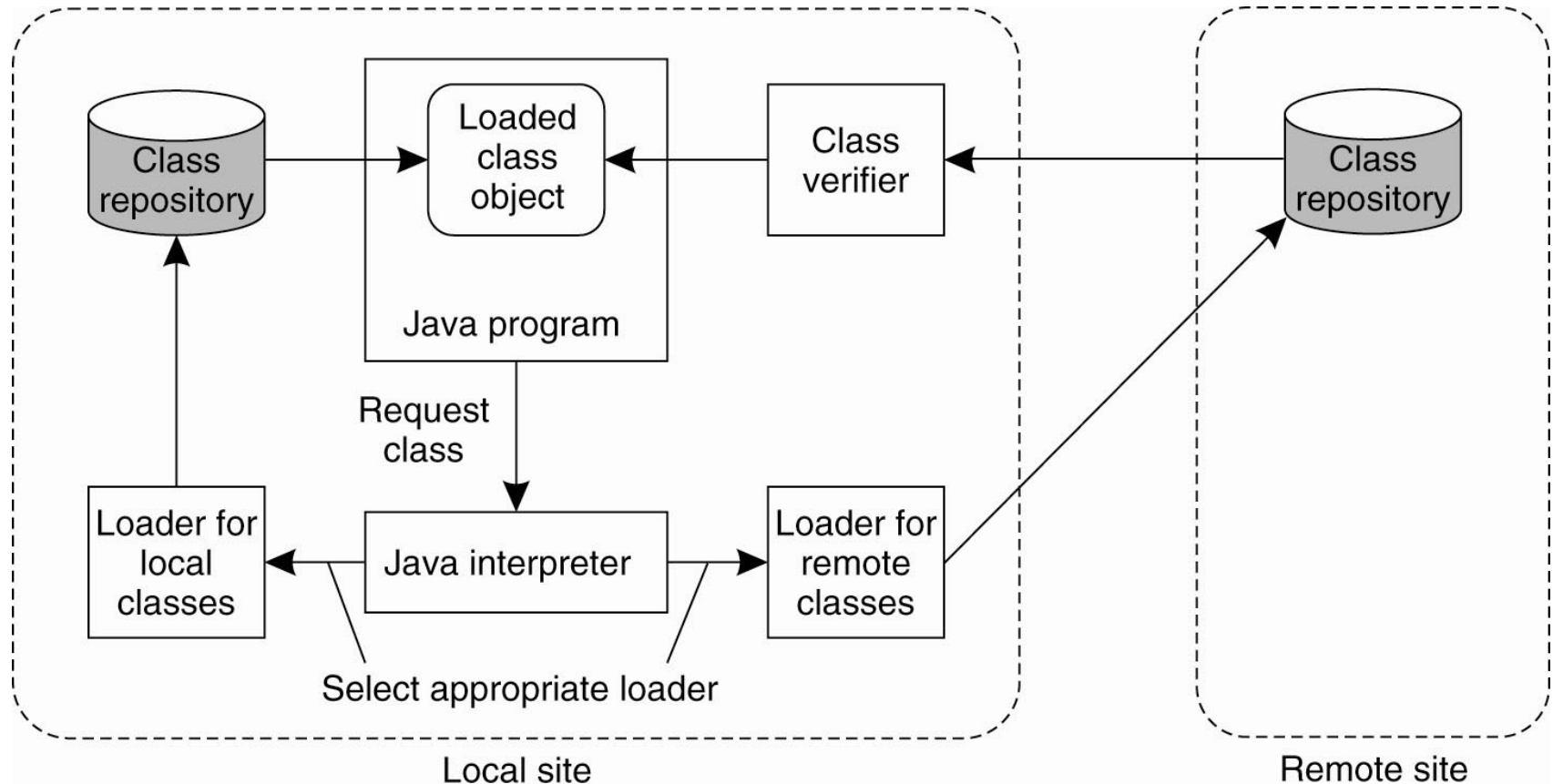
- The hierarchical organization of protection domains as groups of users.



Firewalls



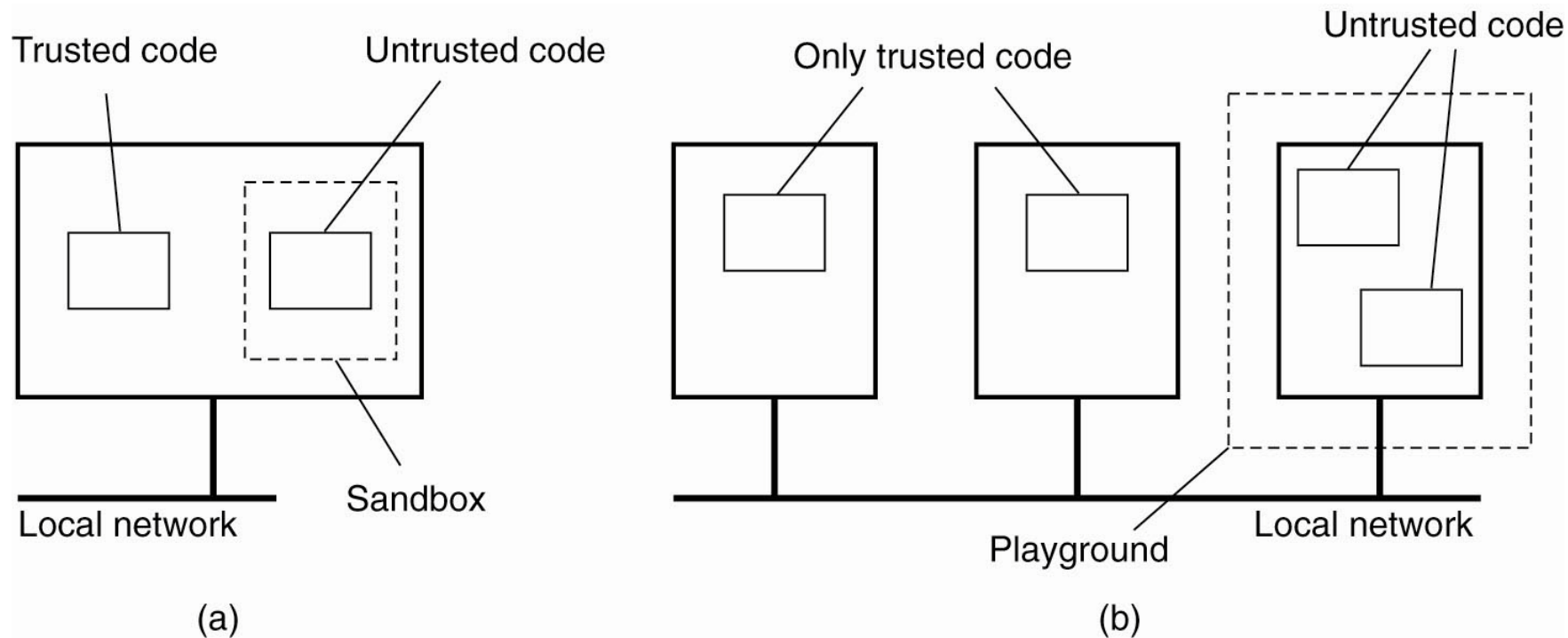
Protecting the Target (1)



- The organization of a Java sandbox.

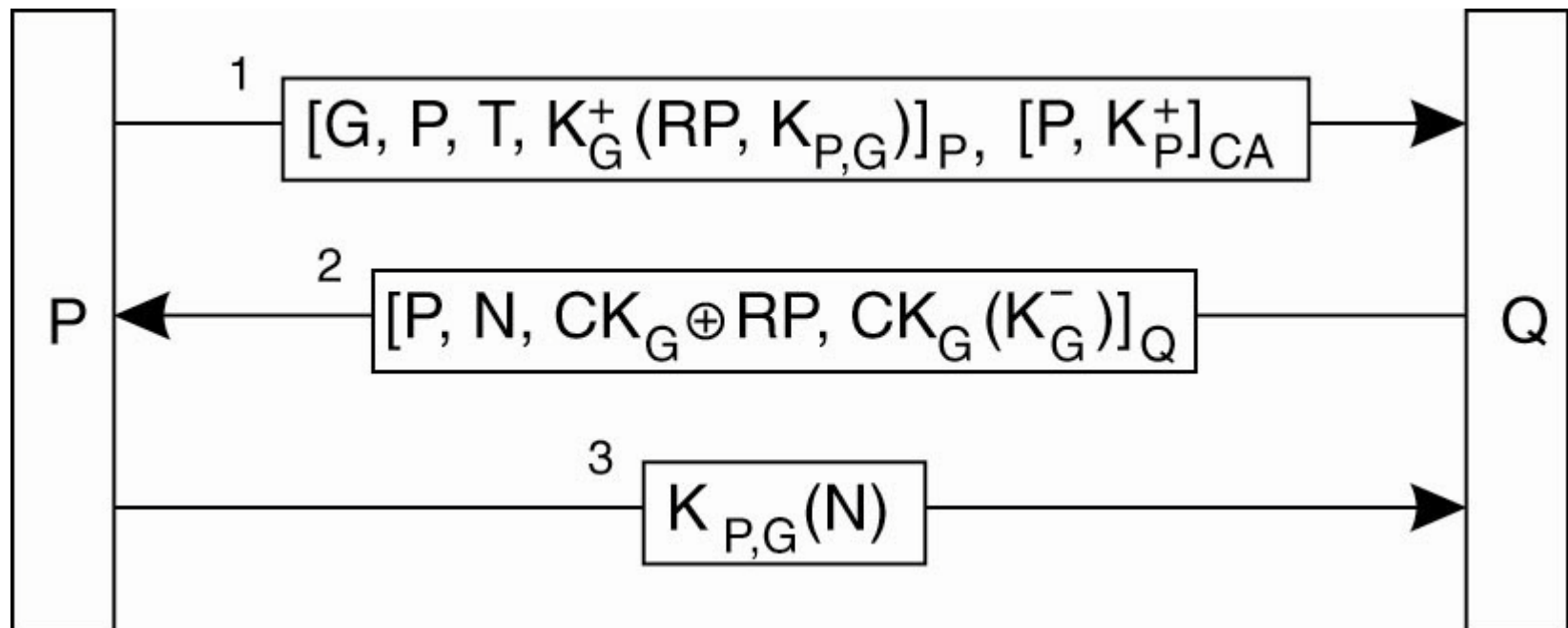
Protecting the Target (2)

- (a) A sandbox. (b) A playground.

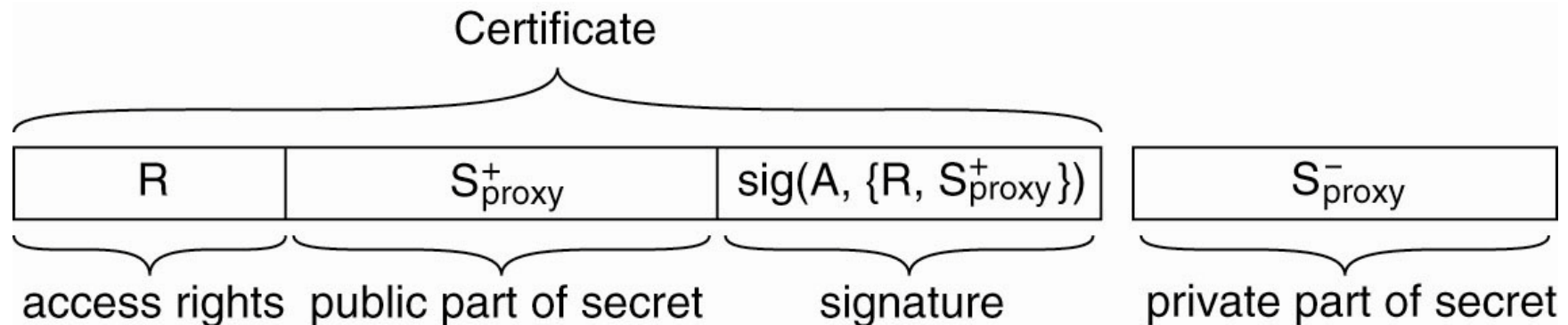


Secure Group Management

- Securely admitting a new group member.

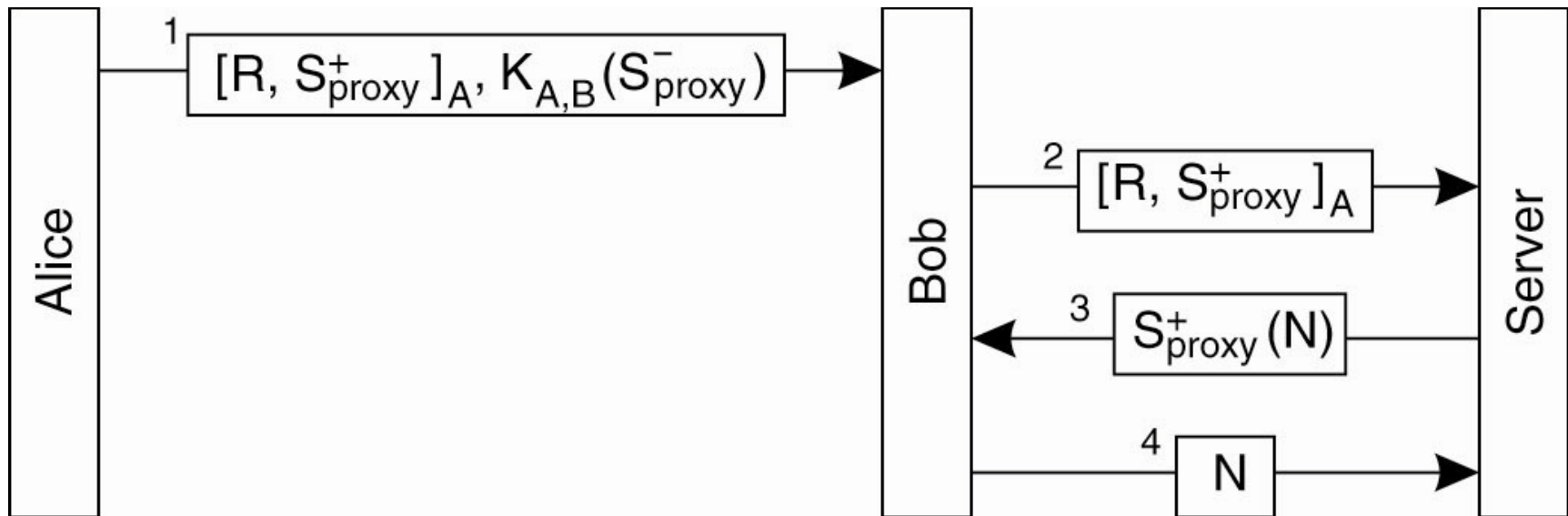


Delegation (1)



- The general structure of a proxy as used for delegation.

Delegation (2)



- Using a proxy to delegate and prove ownership of access rights.